

Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene

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The anomalously large superfluid response of magic-angle twisted bilayer graphene is often discussed in terms of quantum geometry, but the extent to which the orbital structure of the superconducting gap leaves a separable response signature remains less directly quantified. We address this question with an executable Bistritzer-MacDonald-Bogoliubov-de Gennes workflow that tests orbital-projected interband-pairing directions in a finite-band response calculation. The pairing family is defined in the layer-sublattice orbital basis and then projected into the normal-state band basis using a fixed time-reversal-sewn valley convention, allowing the pairing direction and response convention to be varied independently. For the working direction $M_1 = \tau_x \sigma_x$, the projected interband pairing weight increases smoothly with the tuning parameter η and reaches approximately 0.108 at $\eta = 1$ for the $n_{\text{keep}} = 6$ truncation. Dense scans over chemical potential and pairing interpolation show that, at fixed Frobenius norm of the orbital pairing matrix, the normalized isotropic response changes by approximately +0.56% to +11.86% between $\eta = 0$ and $\eta = 1$ over $\mu = -5, \dots, 5$ meV. The same scan at fixed raw gap amplitude is much weaker and can be negative. We therefore identify the signal as a normalization-conditioned mechanism signature rather than an unconditional enhancement of the absolute stiffness. An absolute-unit audit of a previous PRB-style superfluid-weight benchmark shows that the legacy values can be only partially reconstructed from one unified convention. The resulting manuscript policy is that normalized response maps carry the main mechanism claim, while absolute stiffness comparisons are deferred to a separate benchmark calculation with a fully declared convention.

I. INTRODUCTION

Magic-angle twisted bilayer graphene (MATBG) provides a setting in which superconductivity, topology, flat bands, and quantum geometry meet in a single experimentally accessible platform [1–3]. In conventional superconductors the superfluid weight is tied to band dispersion and carrier density, but in flat or nearly flat bands a geometric contribution associated with the quantum metric can remain finite even when ordinary band velocity is small [4, 5]. This geometric contribution is especially relevant for MATBG, whose fragile topology and $C_{2z}\mathcal{T}$ symmetry protect nontrivial band geometry in the flat-band manifold [6–8].

Circuit-QED measurements have reported a superfluid stiffness in MATBG that is large compared with a simple BCS estimate [9]. More recent transport, microwave, and radio-frequency Josephson-junction measurements have directly accessed superfluid stiffness or condensate dynamics and point to nontrivial low-temperature gap structure [10, 11]. Related twisted-trilayer stiffness measurements reinforce the broader nodal-gap phenomenology in magic-angle graphene superconductors [12]. Several theoretical works have connected this anomaly to quantum geometry, remote-band effects, and beyond-BCS corrections [13–15]. Recent dielectric-environment measurements and strong-correlation modeling further emphasize that the microscopic pairing mechanism in

twisted bilayer graphene is not yet settled [16, 17]. In parallel, band-off-diagonal superconductivity has been proposed as a candidate state in twisted graphene superlattices, and its Eliashberg superfluid-stiffness signatures have recently been analyzed [18, 19]. Yet a practical question remains: if a mean-field pairing ansatz contains interband structure after projection into the MATBG band basis, does that structure leave a robust and separable signature in the computed superfluid response?

This paper addresses that question with an explicitly executable finite-band workflow. Rather than beginning from a band-basis off-diagonal gap chosen by hand, we define a pairing matrix in the layer-sublattice orbital basis and project it into the normal-state band basis. This construction makes the pairing direction covariant under band-basis gauge changes and separates two issues that are often entangled: the definition of interband pairing and the response convention used to report stiffness. The main output is a normalized mechanism map,

$$\delta D_{\text{iso}}(\mu, \eta) = \frac{D_{\text{iso}}(\mu, \eta)}{D_{\text{iso}}(\mu, 0)} - 1, \quad (1)$$

computed for two gap-normalization controls.

The central result is deliberately conservative. At fixed Frobenius norm of the orbital pairing direction, increasing the interband-pairing component produces a reproducible change in the isotropic response across chemical potential. At fixed raw gap amplitude, the effect is much smaller. Thus the present calculation supports a normalization-conditioned mechanism signature, not a universal statement that interband pairing always enhances the physical superfluid stiffness. We also per-

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form a self-audit of the absolute units and a previous PRB-style benchmark table; that audit sets the observable policy used below. Normalized response ratios are the claim-bearing quantities in this mechanism paper, whereas absolute stiffness tables should be regenerated from a single declared convention before being used for final experimental comparison.

II. MODEL AND PAIRING CONSTRUCTION

A. Bistritzer-MacDonald model

We use a single-valley Bistritzer-MacDonald continuum Hamiltonian with momenta in $\text{\AA}^{-1}$ and energies in meV [20]. The basis for each moire reciprocal vector $\mathbf{G}$ is

$$(tA, tB, bA, bB), \quad (2)$$

where t, b label top and bottom layers and A, B label sublattices. Unless otherwise stated, the numerical scans use

$$\begin{aligned} \theta &= 1.05^\circ, & v_F &= 2.135 \text{ eV \AA}, \\ w_0 &= 87.2 \text{ meV}, & w_1 &= 109.0 \text{ meV}, \end{aligned} \quad (3)$$

with a square reciprocal cutoff $N_{\text{shell}} = 3$, giving a 196 $\times$ 196 normal-state Hamiltonian per valley and spin. The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4 \text{ \AA}^2, \quad G_M = 0.054047 \text{ \AA}^{-1}. \quad (4)$$

B. Orbital-defined pairing family

The pairing matrix is first specified in the orbital basis. We write

$$\Delta_{\text{orb}}(\eta) = \Delta_0 \mathcal{N}_\eta (M_0 + \eta M_1), \quad (5)$$

with

$$M_0 = \tau_0 \sigma_0, \quad M_1 = \tau_x \sigma_x. \quad (6)$$

Here τ_i act in layer space and σ_i act in sublattice space. The parameter η interpolates between the reference orbital-singlet direction and the interlayer-sublattice-mixed direction. The normalization $\mathcal{N}_\eta$ is treated as a control variable. We use two conventions:

$$\text{fixed Frobenius:} \quad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \quad (7)$$

$$\text{fixed } \Delta_0 : \quad \mathcal{N}_\eta = 1. \quad (8)$$

At each momentum, the orbital pairing is projected into the retained band subspace as

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_-^*(-\mathbf{k}). \quad (9)$$

The working valley convention is a time-reversal-sewn proxy, $U_-(-\mathbf{k}) = U_+^*(\mathbf{k})$. Thus the production scans use $\Delta_{\text{band}} = U_+^\dagger \Delta_{\text{orb}} U_+$ after the valley basis is sewn. A raw independently diagonalized valley-minus basis is not used for W_{inter} because it is not automatically in this sewing convention. This is used as a controlled diagnostic convention rather than a claim that all valley-basis subtleties are solved. The projected interband-pairing weight is

$$W_{\text{inter}} = \frac{\sum_{n \neq m} |\Delta_{nm}|^2}{\sum_{n,m} |\Delta_{nm}|^2}. \quad (10)$$

III. FINITE-BAND RESPONSE

For each retained band subspace we construct a finite-band BdG Hamiltonian,

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\text{band}}(\mathbf{k}) \\ \Delta_{\text{band}}^\dagger(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \quad (11)$$

The static current-current response is evaluated at zero temperature as

$$\Lambda_{\alpha\beta} = \sum_{a \neq b} \frac{\langle a | J_\alpha | b \rangle \langle b | J_\beta | a \rangle (f_a - f_b)}{E_b - E_a}. \quad (12)$$

The values reported by the executable response engine are finite-band raw diagnostics. They are internally consistent response outputs, but they are not used here as final experimental stiffness values.

The band-basis velocity is decomposed into diagonal and off-diagonal parts,

$$v_\alpha^{\text{band}} = v_\alpha^{\text{intra}} + v_\alpha^{\text{inter}}. \quad (13)$$

This yields conventional, geometric, and cross response channels. The isotropic response used in the mechanism scans is

$$D_{\text{iso}} = \frac{1}{2}(D_{xx} + D_{yy}). \quad (14)$$

The numerical checks track BdG Hermiticity, particle-hole symmetry, and exact closure of the channel decomposition.

IV. PAIRING GATES AND RESPONSE MAPS

A. Pairing and velocity gates

The first algebraic pairing gate verifies that an orbital-defined pairing family transforms covariantly under band-basis gauge changes. The BM projection gate then selects $M_1 = \tau_x \sigma_x$ as a useful working direction because it produces a finite projected interband weight in the time-reversal-sewn valley convention while keeping $W_{\text{inter}} = 0$

for $\eta = 0$. For $n_{\text{keep}} = 6$, $nk = 7$, and $\eta = 1$, the dense scan gives

$$W_{\text{inter}} \simeq 0.108. \quad (15)$$

The velocity gate confirms that projected velocity matrices are Hermitian to machine precision and that the intra/inter velocity split closes numerically.

B. Dense chemical-potential and pairing scan

Table I summarizes the dense scan at $nk = 7$ and $n_{\text{keep}} = 6$. The first column gives the retained-band filling proxy ν_{proxy} ; this is a diagnostic occupancy proxy and is not used as a calibrated experimental filling. The Supplemental Material gives a central two-flat-band crosswalk; in that convention the same chemical-potential window spans approximately $\nu_{\text{flat}} = -4$ to $+4$. A filling-sufficiency audit verifies that this crosswalk is adequate as a BM central-flat-band counting reference for the present mechanism maps, but not as a device-level carrier-density calibration.

Figure 1 shows the fixed-Frobenius dense response map. The enhancement is positive over most of the scanned chemical-potential range, with the largest endpoint value in the present scan occurring near $\mu = 5$ meV. The middle panel shows that the geometric fraction remains a useful diagnostic of how the response is redistributed between current sectors.

Figure 2 shows the same scan at fixed raw Δ_0 . The response is much weaker and can be negative. This contrast is the main reason that the conclusion is stated as normalization-conditioned.

C. $nk = 9$ validation

The key-point validation in Table II repeats selected values at $nk = 9$. The fixed-Frobenius signal remains positive and of comparable size, while the fixed- Δ_0 response stays close to zero. Additional selected $nk = 11$ and $nk = 13$ key-point checks, reported in the Supplemental Material, preserve the positive fixed-Frobenius response at the same chemical potentials. A finite-grid trend audit of these key points versus $1/nk^2$ gives positive fixed-Frobenius intercepts between 8.65% and 10.58%, while the fixed- Δ_0 control remains weak and sign-sensitive on the measured grids. The Supplemental Material also reports a targeted $nk = 15$ spot check at $\mu = 0$ and 2 meV, where the fixed-Frobenius response remains +8.97% and +8.65%, respectively. A convergence-sufficiency self-audit combines these selected-grid, trend, and spot-check data. All audit gates pass for the selected-grid normalized mechanism claim, while the same audit explicitly does not promote the numbers to a strict continuum-limit value.

V. SELF-AUDIT OF ABSOLUTE UNITS

The raw response engine outputs values in finite-band diagnostic units. A conservative conversion maps raw $\text{meV } \text{\AA}^2$ to $\text{eV } \text{\AA}^2$ per flavor by dividing by 1000. Any spin/valley degeneracy factor is left outside this conversion:

$$D_{\text{per flavor}}[\text{eV } \text{\AA}^2] = D_{\text{raw}}[\text{meV } \text{\AA}^2]/1000. \quad (16)$$

The moire area and reciprocal scale agree with the standard MATBG geometry, but the prior band-basis decomposition benchmark table [21] is not reproduced by a single global prefactor or by the present production convention. The claim-bearing observable policy for the present manuscript is therefore: the production τ_z/τ_0 finite-band response convention is used for normalized ratios and channel diagnostics, while $\text{eV } \text{\AA}^2$ per-flavor values are restricted to benchmark provenance and self-audit tables.

Table III summarizes the reconstruction audit for uniform s -wave pairing at $\mu = 0$, $\Delta_0 = 1$ meV, and $nk = 14$. The best simple unified reconstruction is

$$D_{\text{conv}} = 2D_{\text{intra}, \tau_z}, \quad D_{\text{geom}} = D_{\text{inter}, \tau_z}. \quad (17)$$

It nearly reconstructs the extended-band endpoint but not the flat-band endpoint. A separate flat-band audit shows that the old $n_{\text{keep}} = 2$ total and conventional values can be approximated with a Gamma-centered mesh and a full curvature-like conventional correction, but that correction does not work cleanly at $n_{\text{keep}} = 6$.

This self-audit changes the interpretation of absolute stiffness values. The historical table is useful as a benchmark, but it should not be used to set the convention for the present interband-pairing mechanism scans. If a later version adds direct experimental comparison, absolute tables should be regenerated from one declared response convention, with degeneracy, Brillouin-zone normalization, and any diamagnetic or curvature term stated explicitly.

VI. DISCUSSION

The main physical message is that an interband-pairing direction can leave a detectable imprint on the normalized finite-band response of MATBG. The imprint is reproducible across chemical potential and survives a modest increase of the momentum grid, but it depends strongly on how the orbital gap direction is normalized. This is both a result and a warning: interband pairing is not a scalar knob. Changing the orbital structure can also change the total norm of the projected gap, and the resulting response should not be interpreted without specifying the normalization convention.

The self-audit also clarifies the status of absolute stiffness claims. The continuum BM model has subtleties associated with UV regularization, current convention, and band truncation. Those subtleties are manageable,

TABLE I. Dense μ scan comparing $\eta = 1$ to $\eta = 0$. The fixed-Frobenius column reports $D_{\text{iso}}(\eta = 1)/D_{\text{iso}}(\eta = 0) - 1$. The fixed- Δ_0 column reports the same quantity without rescaling the orbital pairing matrix.

μ (meV)	ν_{proxy}	fixed Frobenius	fixed Δ_0
-5	-1.483	0.0056	-0.0510
-4	-1.333	0.0891	0.0291
-3	-1.061	0.0785	-0.0071
-2	-0.857	0.0537	-0.0110
-1	-0.653	0.0802	0.0093
0	-0.381	0.0817	0.0112
1	-0.054	0.0914	0.0044
2	0.122	0.0949	0.0119
3	0.395	0.0653	0.0004
4	0.667	0.0662	-0.0005
5	0.857	0.1186	0.0312

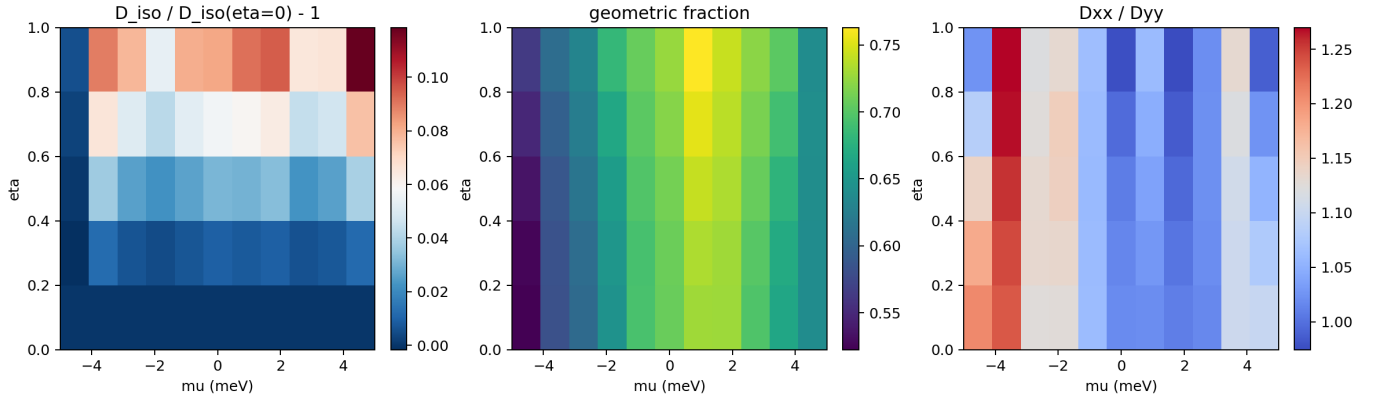


FIG. 1. Dense μ - η response map in the fixed-Frobenius normalization. Left: normalized isotropic response relative to $\eta = 0$ at fixed μ . Middle: geometric fraction. Right: response anisotropy D_{xx}/D_{yy} .

TABLE II. $nk = 9$, $n_{\text{keep}} = 6$ validation of the $\eta = 1$ response.

μ (meV)	fixed Frobenius	fixed Δ_0
-4	0.0604	-0.0068
0	0.0863	0.0103
2	0.0643	-0.0029
4	0.0605	-0.0017

but the audit shows that old absolute tables can mix conventions in ways that are not captured by one simple prefactor. For this reason the most robust near-term observable in the present work is a normalized response map, not an absolute stiffness prediction. Likewise, ν_{proxy} in the tables is a retained-band occupancy proxy defined for scan organization, not a calibrated carrier density. The Supplemental Material gives a central two-flat-band filling crosswalk for the dense scan, but the present figures should still be read as functions of chemical potential and retained-band occupancy, not as a quantitative map onto an experimental supercon-

ducting dome. Within that restricted use, the filling-sufficiency audit shows that the dense scan spans the full central two-flat-band counting range and samples reference points near $\nu_{\text{flat}} = 0$ and ± 2 . The Supplemental Material also includes a claim-scope audit that separates the allowed normalized-response mechanism claims from deferred absolute-stiffness, device-calibration, and continuum-limit claims.

Several limitations remain. First, the valley sewing convention used here is a controlled time-reversal proxy; a fully independent two-valley implementation would be required for valley-asymmetric perturbations or final absolute production calculations. Second, a direct experimental-comparison version would require a final absolute response convention; the present mechanism paper therefore makes no claim-bearing absolute stiffness prediction. Third, the filling variable is currently a retained-band occupancy proxy. Fourth, no self-consistent gap equation is solved. Fifth, a UV-complete lattice calculation would be required for final quantitative comparison with experiment. Recent direct imaging of interaction-reshaped flat bands in MATBG also suggests

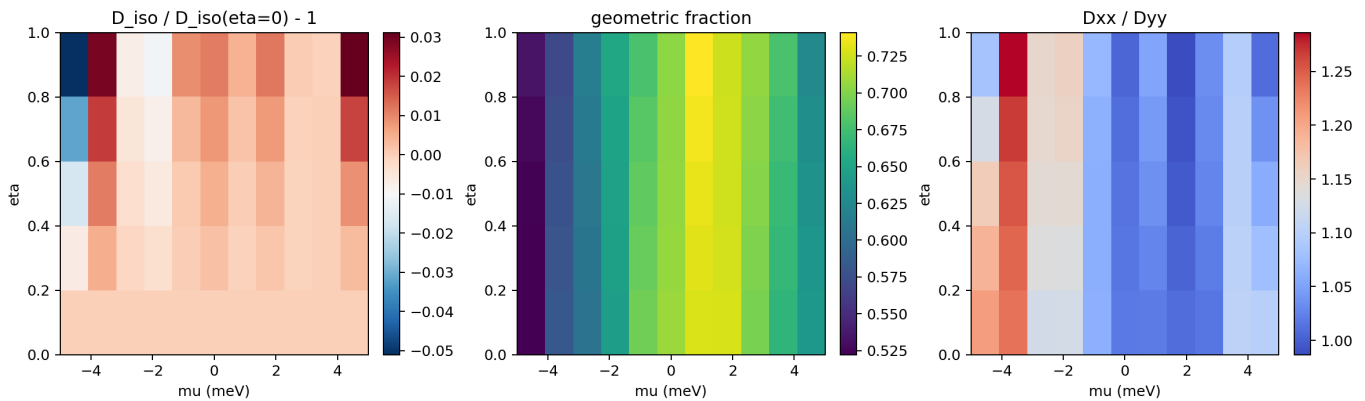


FIG. 2. Dense μ - η response map in the fixed- Δ_0 normalization. Compared with Fig. 1, the normalized response is strongly suppressed and changes sign in parts of the scan.

TABLE III. Prior band-basis benchmark reconstruction. Values are $\text{eV } \text{\AA}^2$ per flavor. The reconstruction route is kept separate from the production interband-pairing scans.

case	n_{keep}	D_{total}	target	D_{conv}	target	D_{geom}
current τ_z/τ_0	2	29.34	67.50	20.95	53.00	8.40
double-conv all- τ_z	2	54.40	67.50	41.89	53.00	12.50
flat endpoint audit	2	66.18	67.50	53.83	53.00	12.36
double-conv all- τ_z	6	126.66	129.30	51.45	54.00	75.22

that a future quantitative comparison should revisit the normal-state band structure beyond the noninteracting BM spectrum used here [22]. Finally, the present ansatz is deliberately simpler than recent microscopic band-off-diagonal and finite-momentum scenarios. Recent tunneling and STM work in magic-angle graphene reports nodal-gap, separated-gap, and intervalley features, while theory has connected band-off-diagonal and Kekulé pair-density-wave structures to spectral and stiffness signatures [18, 19, 23–26]. Those states are complementary to the present uniform orbital-projected diagnostic and will require a separate finite-momentum BdG extension.

VII. CONCLUSIONS

The calculation isolates a concrete finite-band diagnostic of pairing structure in MATBG. For the orbital direction $M_1 = \tau_x \sigma_x$, the projected interband weight is finite and the fixed-Frobenius response map shows a reproducible positive change across the scanned chemical-potential window. The fixed- Δ_0 control, by contrast, is weak and sign-sensitive. This contrast is the central result: interband pairing produces a normalization-conditioned mechanism signature in the finite-band MATBG superfluid response, not an absolute stiffness prediction independent of how the gap direction

is normalized.

The self-audit of units and filling fixes the scope of this conclusion. Normalized response ratios are robust claim-bearing observables for the present mechanism paper, while absolute stiffness tables, device-level filling calibration, and continuum-limit extrapolation remain separate tasks. Within that boundary, the work provides an executable route for comparing orbital gap structure, band projection, and quantum-geometric response in magic-angle graphene superconductivity.

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